What Contextual Holonomy Detects: the All-versus-Nothing Sector, Two Gap Certificates, and the Section Dependence of Wigner–Friend Holonomy

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Abstract

The slogan "contextual holonomy encodes quantum contextuality" is attractive and, as usually stated, false. We give it a sharp, machine-verified form. On the context stack $[X/\mathrm{PU}(n)]$ of maximal abelian subalgebras the canonical degree-2 class has order exactly n; its cocycle holonomy along closed context loops detects *exactly* the all-versus-nothing (AvN) sector of contextuality for Weyl context families (Theorem 1), a two-directional equivalence stress-tested on 2,100 random families at $d = 2, 4$ with zero mismatches. Both containments are strict: at $d = 3$ the class is nontrivial of order 3 while the achievable state-independent AvN value is 0 (Proposition 1, exhaustive certificate), and KCBS exhibits state-dependent contextuality where the class shadow vanishes (Proposition 2). Finally, the holonomy of the Wigner–friend loop generated by H and S splits by level: its projective (Pancharatnam) holonomy is $+1$, while its determinant-section holonomy is exactly $-i$, a τ-level invariant of the 2-adic tower (Observation 1). Interpretational claims must therefore specify loop and level. We state the residual equivalence problem precisely and give hardware-testable predictions.

1 Setting and the repaired slogan

Let X be the space of maximal abelian $*$-subalgebras (MASAs) of $M_n(\mathbb{C})$, $n \geq 3$. The unitary group acts by conjugation; on $\mathrm{U}(n) \ltimes X$ the $\mathrm{U}(1)$-lifting class is vacuous, and after Morita reduction the canonical class on $\mathrm{PU}(n) \ltimes X$ has order exactly n [1, Thm. 1, Thm. 8] (`thmG_general.py`). Weyl context families and their commutator-carry invariant Q are as in [2]. Throughout, "verified" means: reproduced by the named script in the companion repository [5]; expected one-line outputs are pinned in `verification/INDEX.md`.

The repaired slogan is: *the class detects the AvN sector of contextuality, no more and no less.* The next three results make each clause exact.

2 The equivalence on the AvN sector

Fix $d \geq 2$, $m \geq 1$, and a Weyl context family F whose contexts are commuting triples multiplying to a central element. Write the value-assignment system of F as $A\gamma \equiv s \pmod{d}$, where A

is the context–observable incidence matrix and s collects the matrix-computed context phase exponents under the pinned Weyl convention.

Theorem 1 (Equivalence, AvN sector)**.** *F admits no noncontextual value assignment over $\mathbb{Z}_d$ if and only if some cycle of F carries odd anomaly self-pairing Q.*

The criterion direction is [2, Thm. 4]. The solvability reformulation above is verified two-directionally in `holonomy_vs_solvability.py`: over 1,500 random families at $d = 2$ and 600 at $d = 4$, unsolvability of $A\gamma \equiv s$ (decided by exhaustive dual search) coincided with the odd-Q cycle test in every instance, with the Peres–Mermin square ($d = 2$) and the certified $d = 4$ family of [2] pinned as positives so that both truth values of the equivalence are exercised at both d. The achievable obstruction values themselves are classified as $\{0, d/2\}$ for even d and $\{0\}$ for odd d [2, Thm. 1].

3 Both containments are strict

Proposition 1 (Nontrivial class, zero AvN)**.** *On $\mathbb{Z}_3^2$ the Heisenberg cocycle $\omega(u, v) = \zeta_3^{\langle u,v\rangle}$ and its square are coboundaries for none of the $3^9 = 19{,}683$ cochains; the class has order exactly 3. Simultaneously, every cycle of every sampled qutrit Weyl family has even self-pairing, and the achievable AvN value at $d = 3$ is 0.*

Verified exhaustively in `d3_gap_certificate.py`. Hence a nontrivial canonical class does not imply operational (AvN) contextuality: the weakening adopted for the gerbe-theoretic program [3] is necessary, not cautious.

Proposition 2 (Contextuality, zero class shadow)**.** *The KCBS pentagon in $d = 3$ attains quantum value $\sqrt{5} \approx 2.2361$ against the exhaustive classical bound 2, while the state-independent class shadow at $d = 3$ is 0.*

Verified in `kcbs_converse.py`. Hence state-dependent contextuality can be invisible to the class: the converse direction of the naive slogan also fails.

4 The Wigner–friend loop is level-split

Consider the friend-qubit loop $T \to T_x \to T_y \to T$ generated by $U_1 = \nVdash \otimes H$, $U_2 = \nVdash \otimes S$.

Observation 1 (Section dependence)**.** *The projective (Pancharatnam) holonomy of the loop is $+1$: the two eigenvector strands carry Bargmann phases $e^{\pm i\pi/4}$ (the Bloch-octant $-\Omega/2$ law), whose product is 1, gauge-invariantly. The determinant-section holonomy is exactly $-i$ for every special-unitary closing arrow; $-i = \tau^{-1}$ at $d = 2$, a τ-level invariant of the 2-adic tower, with section-independent square -1.*

Verified in `wf_loop_holonomy.py` (200 gauge randomizations; 500 closers). Any physical or interpretational claim about "the" holonomy of this loop must state the level; the two answers differ.

Hardware predictions. The strand phases $\pm 45°$ and loop product $+1$ are interferometrically measurable with two-qubit Hadamard tests; on the same device the Peres–Mermin loop yields context signs $(+, +, +, +, +, -)$, i.e. ray-level holonomy -1. One machine, two loops, opposite holonomies, both sharp.

Measured (`ibm_fez`, 2026-07-09, job `d986q62f47jc73a7hm2g`; registry record `hardware/results/ibm_fez_hol` strand phases $+45.88°$ and $-48.81°$ against targets $\pm 45°$ (per-strand statistical error $\approx 1.1°$; the $\sim 3.5\sigma$ offset on the X-prepared strand is a plausible preparation systematic), loop phase $-2.93° \pm 1.55°$ against target $0°$. The ray-level holonomy of the Wigner–friend loop is thus measured to be $+1$; the value $-i$ (loop $-90°$) is excluded at this level by $\approx 56\sigma$, in accordance with Observation 1: $-i$ is a section-level, not interferometric, invariant. On the same device the Peres–Mermin loop gave context products $(+0.750, +0.835, +0.722, +0.780, +0.824, -0.700)$, sign pattern 6/6 as predicted, and $S = 4.6125 \pm 0.0173$: a 35.4σ violation of the exact deterministic noncontextual bound 4 (the run-time script carried the loose $n_c - 1 = 5$ bound, since replaced in `qkernel` by the exhaustive tight bound), consistent with the archived `ibm_fez` value 4.558 ± 0.018.

5 The residual problem

Open Problem 1. *Characterize, in invariants of the context stack $[X/\mathrm{PU}(n)]$ refined beyond the canonical H^2 class, the property "some state has positive contextual fraction on F". Equivalently: identify the cohomological home of KCBS-type (state-dependent, class-invisible) contextuality, and prove or refute that a sheaf-theoretic refinement restores a two-directional equivalence.*

Propositions 1–2 show no formulation at the level of the canonical class alone can succeed; Theorem 1 shows the AvN sector is already optimally captured there.

Conjecture 1 (State-sector pairing). *For odd prime d, the state-dependent sector is classified by the pairing of the state against the order-d class through the Weyl transform: a state yields positive contextual fraction on stabilizer measurement scenarios if and only if its parity-frame quasi-distribution is negative — the Howard–Wallman–Veitch–Emerson contextuality–negativity correspondence [4], recast as a stack pairing. For even d, where no Gross frame exists, the corresponding role is played by the top layer of the 2-adic tower of [2].*

Computed anchors (`state_sector_probe.py`): CF(PM) $= 1.0000$ for every state (AvN); CF(KCBS) $= 0.4721 = 2\sqrt{5} - 4$ at the optimal state with class shadow 0; at $d = 3$ the parity-frame quasi-distribution is nonnegative on all twelve stabilizer pure states (exhaustive) and attains $-1/3$ on the strange state. Notably, the τ-convention frame of [2] exhibits $-1/18$ pseudo-negativity on stabilizer states at $d = 3$: τ is the even-d object, and the odd/even divide of [1] resurfaces at the level of quasi-probability frames.

A Claims-to-verification table

Claim	Script	Expected one-liner
Thm. 1	`holonomy_vs_solvability.py`	0 mismatches; pinned PM/cert4 agree
Prop. 1	`d3_gap_certificate.py`	order 3; AvN $\{0\}$; gap certified
Prop. 2	`kcbs_converse.py`	$2.23607 > 2$; converse gap certified
Obs. 1	`wf_loop_holonomy.py`	strands $\pm\pi/4$; det-holonomy $-i$
PM anchor	`pm_gauge_invariance.py`	product -1 over all 512 gauges

References

[1] M. Flores Gordillo, *The Canonical Class of the MASA Context Stack and the Peres–Mermin Obstruction*, working draft, 2026.

[2] M. Flores Gordillo, *The Obstruction Spectrum of Weyl Context Families and the 2-adic Tower*, working draft, 2026.

[3] M. Flores Gordillo, *Contextual Holonomy, Lifting Bundle Gerbes, and the Geometric Structure of Quantum Contextuality*, in preparation.

[4] M. Howard, J. Wallman, V. Veitch, J. Emerson, *Contextuality supplies the magic for quantum computation*, Nature 510, 351 (2014).

[5] `github.com/manuflog/contextuality-obstructions`, verification suite, 2026.